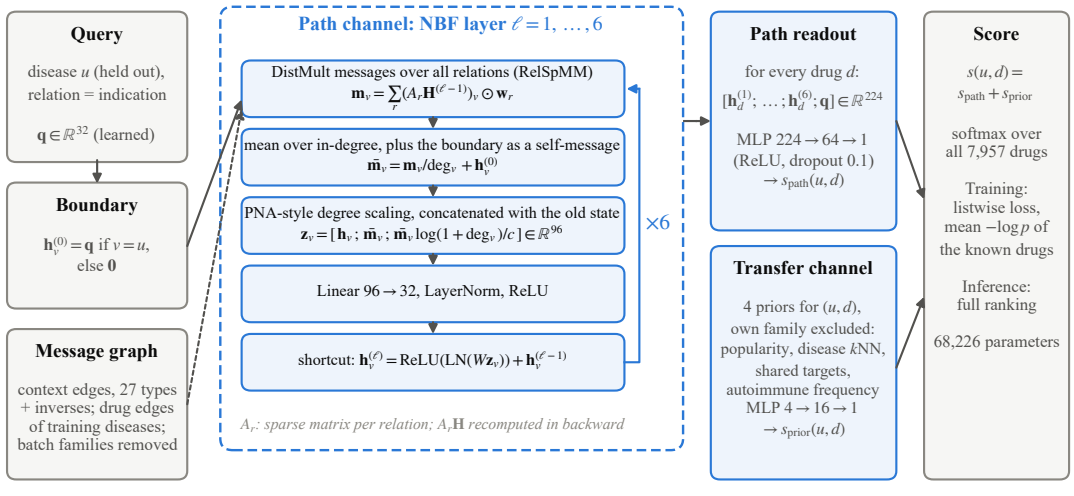

Query
disease u (held out), relation = indication
$\mathbf{q} \in \mathbb{R}^{32}$ (learned)
Boundary
$\mathbf{h}_v^{(0)} = \mathbf{q}$ if $v = u$, else $\mathbf{0}$
Message graph
context edges, 27 types + inverses; drug edges of training diseases; batch families removed
Path channel: NBF layer $\ell = 1, \ldots, 6$
DistMult messages over all relations (RelSpMM)
$\mathbf{m}_v = \sum_r (A_r \mathbf{H}^{(\ell-1)})_v \odot \mathbf{w}_r$
mean over in-degree, plus the boundary as a self-message
$\tilde{\mathbf{m}}_v = \mathbf{m}_v / \deg_v + \mathbf{h}_v^{(0)}$
PNA-style degree scaling, concatenated with the old state
$\mathbf{z}_v = [\mathbf{h}_v ; \tilde{\mathbf{m}}_v ; \tilde{\mathbf{m}}_v \log(1+\deg_v)/c] \in \mathbb{R}^{96}$
Linear $96 \to 32$, LayerNorm, ReLU
shortcut: $\mathbf{h}_v^{(\ell)} = \mathrm{ReLU}(\mathrm{LN}(W\mathbf{z}_v)) + \mathbf{h}_v^{(\ell-1)}$
×6
A_r: sparse matrix per relation; $A_r\mathbf{H}$ recomputed in backward
Path readout
for every drug d:
$[\mathbf{h}_d^{(1)}; \ldots; \mathbf{h}_d^{(6)}; \mathbf{q}] \in \mathbb{R}^{224}$
MLP $224 \to 64 \to 1$ (ReLU, dropout 0.1)
$\to s_{\text{path}}(u, d)$
Transfer channel
4 priors for (u, d), own family excluded: popularity, disease kNN, shared targets, autoimmune frequency
MLP $4 \to 16 \to 1$
$\to s_{\text{prior}}(u, d)$
Score
$s(u, d) = s_{\text{path}} + s_{\text{prior}}$
softmax over all 7,957 drugs
Training: listwise loss, mean $-\log p$ of the known drugs
Inference: full ranking
68,226 parameters